25901869 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

second SEMESTER

Skill Development Components (SDC) - MG2SDCREM101 - SOLAR PHOTOVOLTAIC SYSTEMS

(2025 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **15** questions

Each question carries **1** marks

1. Write the standard unit for PV module rating. [A] / [CO1]
2. Write one example of second generation solar cell. [A] / [CO1]
3. Write the definition of a solar PV module. [A] / [CO1]
4. Write the function of diode in the equivalent circuit. [A] / [CO1]
5. Associate shading with reduction in PV module power output. [U] / [CO2, CO5]
6. Relate load demand to battery capacity in a standalone PV system. [U] / [CO2]
7. Write one purpose of a multi-branch connector. [A] / [CO1]
8. Classify the effect of shading on a series-connected PV string. [U] / [CO2]
9. Show the symbol for a series connection. [A] / [CO1]

10. Select one method to ensure stable operation of a grid-connected PV system. [A] / [CO3]

11. Define overcurrent protection in the context of PV Combiner Boxes. [K] / [CO4]

12. Recognize the effect of loose connections inside a junction box on system performance. [K] / [CO4]

13. Explain why insulation resistance decreases with moisture (in one sentence). [U] / [CO4]

14. Select the device used for isolating the AC supply in ACDB. [K] / [CO4]

15. Relate temperature rise to reduction in wire current carrying capacity. [U] / [CO4]

16. Write one electrical factor affecting PV module efficiency. [A] / [CO1]

17. Relate module voltage rating to battery voltage in a standalone system. [U] / [CO2, CO5]

[1x15 = 15]

Part B

Short Essay

Answer any **3** questions

Each question carries **5** marks

18. Illustrate the contrast between maximum power point and open circuit condition using the I–V characteristics of a solar cell. [A] / [CO1]

19. Present the comparison between the function of a bypass diode and the condition when no bypass diode is used in a PV module. [A] / [CO1]

20. Write the advantages and limitations of lithium-ion batteries compared to lead-acid batteries for solar applications. [A] / [CO3]

21. Explain the factors affecting voltage drop in DC cabling [U] / [CO4]

22. Classify the electrical and physical parameters of a PV module and explain their importance in system design. [A] / [CO1]

[5x3 = 15]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

23. Present the differences between field measurement and STC measurement of PV module parameters, including the effect on calculated efficiency and output. [A] / [CO1]

24. Discuss the impact of light intensity and temperature on the output characteristics of a silicon solar cell. [U] / [CO2]

25. Illustrate the proper charging and discharging procedure to maximize battery life. [A] / [CO3]

26. Explain the procedure for designing an effective grounding system for a substation or industrial installation. [U] / [CO4]

[10x2 = 20]